resolving time's arrow also requires sufficiently uncorrelated initial quantum fluctuations. Again, if we think in a Boltzmann-like manner, among all the fluctuations yielding conditions ripe for inflation, sooner or later there will be one that meets this condition as well, allowing the universe as we know it to initiate.

6. There are some physicists who would claim that the situation is better than described. For example, Andrei Linde argues that in chaotic inflation (see note 11, Chapter 10), the observable universe emerged from a Planck-sized nugget containing a uniform inflaton field with Planck scale energy density. Under certain assumptions, Linde further argues that the entropy of a *uniform* inflaton field in such a tiny nugget is roughly equal to the entropy of any other inflaton field configuration, and hence the conditions necessary for achieving inflation weren't special. The entropy of the Planck-sized nugget was small but on a par with the possible entropy that the Planck-sized nugget *could* have had. The ensuing inflationary burst then created, in a flash, a huge universe with an enormously higher entropy—but one that, because of its smooth, uniform distribution of matter, was also enormously far from the entropy that it could have. The arrow of time points in the direction in which this entropy gap is being lessened.

While I am partial to this optimistic vision, until we have a better grasp on the physics out of which inflation is supposed to have emerged, caution is warranted. For example, the expert reader will note that this approach makes favorable but unjustified assumptions about the high-energy (transplanckian) field modes—modes that can affect the onset of inflation and play a crucial role in structure formation.

Chapter 12

1. The circumstantial evidence I have in mind here relies on the fact that the strengths of all three nongravitational forces depend on the energy and temperature of the environment in which the forces act. At low energies and temperatures, such as those of our everyday environment, the strengths of all three forces are different. But there is indirect theoretical and experimental evidence that at very high temperatures, such as occurred in the earliest moments of the universe, the strengths of all three forces converge, indicating, albeit indirectly, that all three forces themselves may fundamentally be unified, and